

AI/NLP Financial Risk Engine & Strategic Portfolio Stress Testing

Module B: Autonomous Multi-Asset Crisis Simulation Engine

CANDIDATE INFORMATION

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Track / Module:

MODULE B (Strategic Portfolio Stress Testing)

SYSTEM HIGHLIGHTS AT A GLANCE

· FinBERT NLP: P(pos) - P(neg) Sentiment Scoring [-1.0 to +1.0]

· 11-Category Financial Event Taxonomy & Calibrated Impact Scoring (1-10)

· Multi-Asset Synthetic Portfolio (INR 100M: Equities, Bonds, Loans, Derivs)

· Autonomous Stress Trigger on Severity > 7.0 with Predefined Asset Shocks

Problem Statement & Solution Approach

THE BUSINESS PROBLEM

Unstructured Financial Noise: Financial news and disclosures arrive at unprecedented velocity across GDELT and NewsAPI feeds.

Lagging Traditional Indicators: Standard financial ratios reflect past performance rather than forward-looking black-swan risk events.

Disconnected Stress Testing: Institutional risk models rarely link incoming geopolitical or credit headlines to immediate multi-asset shocks.

Need for Machine-Readable Signals: Financial analysts need quantified, reproducible severity metrics and automated stress simulation.

OUR FIVE-PHASE SOLUTION APPROACH

Phase 1: Ingests & normalizes news from GDELT & NewsAPI with SHA-256 deduplication and entity-to-ticker mapping.

Phase 2: Fine-tuned ProsusAI FinBERT delivers sentiment P(pos)-P(neg), 11-class event taxonomy, and calibrated 1-10 severity impact scoring.

Phase 3: Integrates public market volatility (yfinance) across 10 benchmark tickers and creates an INR 100M synthetic 4-tier portfolio.

Phase 4 (Module B): Automates stress test execution when Impact > 7.0, applying predefined shocks to Equities, Bonds, Loans, and Derivatives.

Phase 5: Exposes machine-readable FastAPI REST endpoints & a glassmorphic dashboard with interactive breaking-news stress simulation.

System Design & End-to-End Architecture

1. Data Ingestion	2. AI/NLP Risk Engine	3. Risk Intelligence	4. Module B Stress Test
· GDELT 2.0 DOC API · NewsAPI v2 / Everything · Local Sample News Dataset · SHA-256 Deduplication · ISO 8601 Timestamps	· ProsusAI FinBERT Model · P(pos) - P(neg) Score [-1, +1] · 11-Category Taxonomy · Calibrated Impact (1-10) · Composite Confidence	· 10 Mega-Cap Equities · yfinance 30-Day Volatility · INR 100M Multi-Asset Basket · Equities, Bonds, Loans, Derivs · Shock Matrix Predefinition	· Gating: Impact Score > 7.0 · Event-to-Shock Mapping · Asset Haircut Simulations · Pre vs Post Valuation Calc · Loss % & Audit Logging

PERSISTENCE, REST APIS & PRESENTATION LAYER

- SQLite Database (SQLAlchemy): 'articles', 'risk_signals', and 'stress_test_runs' tables provide full reproducibility & zero-setup portability.
- FastAPI Web Server: 10 REST endpoints exposing machine-readable JSON signals, portfolio metrics, custom text analysis, and audit trails.
- Glassmorphism Dashboard: Chart.js interactive charts, real-time stress testing triggers, and live jury custom headline simulator.

Implementation Highlights & Key Tech Choices

FinBERT Sentiment Analysis

Hugging Face / ProsusAI FinBERT

- Trained specifically on financial text corpora (Financial PhraseBank).
- Formula: $\text{Sentiment} = P(\text{positive}) - P(\text{negative})$ bounded in $[-1.0, +1.0]$.
- Captures financial nuances (e.g. 'unhedged default', 'margin calls').
- Resilient fallback ensures zero crashes during offline evaluation.

Calibrated Impact Scoring

Domain-Specific Severity Formula

- Calibrated scale from 1.0 to 10.0 reflecting systemic disruption potential.
- Base category weights: Market Crash (8.5), Bankruptcy (8.2), Geopolitics (7.6).
- Sentiment amplification: Negative sentiment increases downside shock severity.
- Explicit disclaimer: Scenario analysis metric, NOT a market prediction.

Module B Strategic Stress Testing

Multi-Asset Scenario Engine

- Triggered autonomously when AI Impact Score > 7.0 threshold.
- Synthetic Portfolio: Equities (·40M), Bonds (·25M), Loans (·20M), Derivatives (·15M).
- Applies distinct asset-class shock vectors (e.g. Crash: Eq -15%, Deriv -18%).
- Calculates pre/post valuations, rupee losses, and persists audit trail.

Production Architecture

FastAPI + Pydantic + Pytest

- Strict schema validation via Pydantic v2 BaseSettings and BaseModel.
- Comprehensive 28-test automated regression suite covering all 5 phases.
- One-command launcher (python run_app.py) with dual LIVE & DEMO modes.
- Self-contained zero-binary repository with complete public datasets.

Key Results & Stress Testing Simulations

Event Type	Headline Scenario	FinBERT Sent.	Impact	Pre-Stress	Post-Stress	Simulated Loss
MARKET_CRASH	Algorithmic Cascade Flash Crash	-0.85	8.8 / 10	-100.0M	-89.05M	--10.95M (-10.95%)
GEOPOLITICAL	Taiwan Strait Semiconductor Blockade	-0.75	8.2 / 10	-100.0M	-92.35M	--7.65M (-7.65%)
CREDIT_EVENT	Commercial Real Estate Rating Downgrade	-0.78	7.9 / 10	-100.0M	-91.45M	--8.55M (-8.55%)
BANKRUPTCY	Fintech Credit Lender Chapter 11 Run	-0.82	8.1 / 10	-100.0M	-89.50M	--10.50M (-10.50%)
REGULATORY	Antitrust Mandatory Breakup Lawsuit	-0.68	7.5 / 10	-100.0M	-94.45M	--5.55M (-5.55%)
EARNINGS	Apple Record Enterprise Revenue Beat	+0.65	4.5 / 10	-100.0M	-100.0M	-0.0 (Monitoring)
PRODUCT_LAUNCH	Tesla Autonomous Robotaxi Fleet Rollout	+0.55	4.0 / 10	-100.0M	-100.0M	-0.0 (Monitoring)

TESTING ACCURACY & VALIDATION METRICS

- 100% Test Suite Coverage across 28 automated tests (pytest). Zero unhandled runtime exceptions.
- Deterministic, explainable trigger rule: Impact > 7.0 prevents false positive stress runs.

Domain Impact & Financial Relevance

Credit Risk Underwriting

- Early Liquidity Run Warning:
Identifies credit deterioration and margin call pressures before financial statements publish.
- Loan Portfolio Sensitivity:
Enables syndication desks to assess asset write-downs on commercial facilities (·20M exposure).
- Counterparty Contagion Mapping:
Links regional banking stresses directly to derivative counterparty exposures.

Asset & Wealth Management

- Proactive Tail-Risk Hedging:
Gives portfolio managers real-time visibility into post-shock valuations (up to -10.95% tail loss).
- Cross-Asset Stress Correlation:
Moves beyond single-stock betas to systemic multi-asset impact across Equities, Bonds, and Derivatives.
- Explainable Decision Support:
Every stress shock maintains documented simulation assumptions for investment committees.

Regulatory & Capital Adequacy

- Autonomous Scenario Generation:
Automates CCAR / Basel-style scenario construction from live geopolitical & economic bulletins.
- Audit Trail Compliance: Every evaluated headline records Run ID, pre/post valuations, and exact timestamps in SQLite.
- Interactive Jury Simulator:
Allows chief risk officers to test 'What-If' scenarios instantaneously without manual recalculation.

CURRENT ASSUMPTIONS & BOUNDARIES

- 1. Simulation vs Prediction: Impact scores and stress tests represent predefined scenario sensitivity shocks, NOT empirical forecasts.
- 2. Public / Synthetic Data Only: Relies strictly on public news (GDELT, NewsAPI) and synthetic asset portfolios (-100M). Zero proprietary data used.
- 3. Uniform Class Shocks: Asset classes experience uniform percentage shocks rather than security-specific non-linear derivative greeks.
- 4. GDELT Content Truncation: Live GDELT artist queries provide titles and domain snippets rather than full unpaywalled news text.

FUTURE ENHANCEMENTS & ROADMAP

- 1. Real-Time WebSocket Streaming: Push instant alert notifications to trading terminals when breaking events exceed Impact > 7.0 threshold.
- 2. Non-Linear Derivatives Greeks: Incorporate Black-Scholes Delta, Gamma, and Vega recalculations under market crash volatility spikes.
- 3. Knowledge Graph Entity Linking: Connect global supply chain nodes (e.g. TSMC to NVDA/AAPL) for cross-company cascade risk tracking.
- 4. Historical Backtesting Engine: Validate predicted stress losses against actual market drawdowns during 2020 COVID and 2008 Lehman events.